



Medical chatbots in healthcare

Doctors assess disease progression using radiology expert analysis and scan reports. Neural networks and the deep learning algorithms of AI can go a long way in accelerating the speed of the analysis and auto-generating the scan reports. Image-to-image neural networks use the new images (the input layer) with their internally trained disease data sets (hidden layer) to generate output images (output layer). These neural networks work on feedforward and backpropagation mechanisms to thoroughly investigate the raster image data sets.

There are two benefits of adapting to neural networks based disease monitoring. First, conventionally, scanning machines use high radiation exposure in patients to generate high quality or clear scanned images. High-quality scanned images came at the cost of a patient's health. High radiation poses the risk of cancer and it has been considered an inevitable side-effect for those who take scans. However, with image-to-image neural networks, it is now possible to let the patient get scanned with minimum radiation exposure. Low quality captures, distorted images,

or other minuscule errors (termed as data noise) will work. The deep learning algorithms compare the input images with their internally trained mass classifier data sets, to auto reduce all data noise. Neural networks are trained to recognise noise signature distributions, improve image resolutions, enhance image contrasts and provide a more accurate analysis of the disease's progression.

The second benefit is the quality of the prognosis and the accuracy

of the results. Radiology experts no longer need to perform a detailed subjective analysis on every scan or testing sample that comes their way — a task that is time-intensive, repetitive and costly. A neural network can do the same job with greater accuracy (up to 95 per cent) in just a few seconds. With the ongoing COVID-19 pandemic, patients with respiratory illnesses like pneumonia wait to get themselves scanned and see the results. However, a few hospitals implement applications like qXR to calculate the risk of COVID-19 infections using neural networks. Smart AI solutions help them get a quicker diagnosis of disease progression. The swift, timely results also cascade into other healthcare processes, like providing prompt medical interventions and immediate treatment, thereby increasing the likelihood of recovery. For active disease control, especially during pandemics, neural networks can deliver test results in a much more robust and faster way.

Automation for disease tracking

Public health care systems have been around since the last decade. All hospitals in India have automated patient records, but for administration and billing purposes only.



AI and machine learning for disease monitoring